Daily Algebra Drill #5

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Partial fractions with quadratic denominators · Elementary functional equations · Cauchy–Schwarz (Engel form) · Conjugate surd pairs revisited.

Section A Rapid Recognition

1. Simplify $\dfrac{x^2-9}{x^2+5x+6}$.

Ans: $\dfrac{x-3}{x+2}$

2. Rationalise: $\dfrac{4}{\sqrt{7}-\sqrt{3}}$.

Ans: $\sqrt{7}+\sqrt{3}$

3. Given $x+y=3$, $xy=-4$, find $(x-y)^2$.

Ans: 25

4. Factorise: $x^3 - x^2 - 4x + 4$.

Ans: $(x-1)(x-2)(x+2)$

5. Without expanding, find the coefficient of x^2 in $(1+x)^4$.

Ans: 6

6. Write the quadratic with roots α, β where $\alpha + \beta = -3$, $\alpha\beta = -10$.

Ans: $x^2 + 3x - 10 = 0$

7. Simplify: $\sqrt{50} - \sqrt{18} + \sqrt{8}$.

Ans: $4\sqrt{2}$

8. Given $a + b = 7$, $a^2 + b^2 = 29$, find $a^3 + b^3$.

Ans: 133

9. Solve: $|2x - 3| = 7$.

Ans: $x = 5$ or $x = -2$

10. Find the minimum of $\dfrac{(x+4)^2}{x}$ for $x > 0$.

Ans: 16

Section B Manipulation Drills

1. Partial fractions: $\dfrac{5x+1}{(x+1)(x-2)}$.

Ans: $\dfrac{4/3}{x+1}+\dfrac{11/3}{x-2}$, or $\dfrac{3(x+1)}{4}+\dfrac{3(x-2)}{11}$

2. Partial fractions: $\dfrac{3}{x(x^2+3)}$.

Ans: $\dfrac{1}{x}-\dfrac{x}{x^2+3}$

3. Show $\dfrac{1}{\sqrt{n}+\sqrt{n+1}}=\sqrt{n+1}-\sqrt{n}$; hence evaluate $\sum_{k=0}^{8}\dfrac{1}{\sqrt{k}+\sqrt{k+1}}$.

Ans: 3

4. Factorise $a^4 + a^2 + 1$ over $\mathbb{Z}$.

Ans: $(a^2 + a + 1)(a^2 - a + 1)$

5. Express $(a - b)^4$ in terms of $s = a + b$ and $p = ab$.

Ans: $(s^2 - 4p)^2$

6. Simplify $\dfrac{x^2 - 2x + 1}{x^2 + 2x - 3} \div \dfrac{x^2 - 1}{x^2 + 4x + 3}$.

Ans: 1

7. Find the quadratic with roots r^2 and s^2 where r, s are roots of $x^2 + px + q = 0$.

Ans: $x^2 - (p^2 - 2q)x + q^2 = 0$

8. Solve: $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{5}{6}$ and $\dfrac{1}{x} - \dfrac{1}{y} = \dfrac{1}{6}$.

Ans: $x = 2,\ y = 3$

9. Prove: $\dfrac{1}{(2k-1)(2k+1)} = \dfrac{1}{2}\left(\dfrac{1}{2k-1} - \dfrac{1}{2k+1}\right)$; hence $\sum_{k=1}^{n} \dfrac{1}{(2k-1)(2k+1)} = \dfrac{n}{2n+1}$.

Ans: Proof by partial fractions, then telescoping.

10. If $f(x) = \dfrac{ax+b}{cx+d}$ and $f(f(x)) = x$, what condition must a, b, c, d satisfy?

Ans: $a + d = 0$ (equivalently $d = -a$)

Section C Substitution & Structure

1. Solve: $\left(x^2 + \dfrac{1}{x^2}\right) - 7\left(x - \dfrac{1}{x}\right) - 10 = 0$.

Ans: $x - \dfrac{1}{x} = 8$ (giving $x = 4 \pm \sqrt{17}$) or $x - \dfrac{1}{x} = -1$ (giving $x = \dfrac{-1 \pm \sqrt{5}}{2}$)

2. Using the Engel (Titu) form of Cauchy–Schwarz, prove $\dfrac{a^2}{b} + \dfrac{b^2}{c} + \dfrac{c^2}{a} \geq 3$ when $a + b + c = 3$.

Ans: Proved below.

3. Solve the system: $x^2 + xy = 12,\ \ xy + y^2 = 6$.

Ans: $(x, y) = (2\sqrt{2},\ \sqrt{2})$ and $(-2\sqrt{2},\ -\sqrt{2})$

4. Find values of k for which $kx^2 - 4x + k - 3 = 0$ has two distinct real roots.

Ans: $-1 < k < 4$ and $k \neq 0$

5. Simplify $\sqrt{3 + 2\sqrt{2}} + \sqrt{3 - 2\sqrt{2}}$.

Ans: $2\sqrt{2}$

6. If $x - \dfrac{1}{x} = 4$, find $\sqrt{x} - \dfrac{1}{\sqrt{x}}$ exactly.

Ans: $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{2\sqrt{5} - 2}$.

7. Given $a, b > 0$ with $\frac{1}{a} + \frac{1}{b} = 1$, show $(a-1)(b-1) = 1$ and find $\min(a+b)$.

Ans: $(a-1)(b-1) = 1$ always; minimum of $a+b$ is 4, at $a = b = 2$.

8. Solve: $(x^2 - x - 1)^2 = (2x+1)^2$.

Ans: $x = 0,\ x = -1,\ x = \dfrac{3 \pm \sqrt{17}}{2}$

Section D Challenge (SMC / BMO1 difficulty)

1. Find $f(x)$ given $f(x) + f\left(\frac{1}{1-x}\right) = x$.

Ans: $f(x) = \frac{x^3 - x + 1}{2x(x-1)}$

2. Prove the Cauchy–Schwarz inequality in Engel form; find $\min\left(\frac{1}{x} + \frac{4}{1-x}\right)$ for $0 < x < 1$.

Ans: Minimum is 9 at $x = \frac{1}{3}$.

3. Find all $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x + y) = f(x) + f(y)$ and $f(1) = 3$.

Ans: $f(x) = 3x$ for $x \in \mathbb{Q}$; with continuity assumption, $f(x) = 3x$ for all $x \in \mathbb{R}$.

4. Show that among any 5 integers there exist 2 whose difference is divisible by 4. Hence prove that for any 5 integers, two of them satisfy $a^2 \equiv b^2 \pmod 4$.

Ans: Proved below.

5. Find all integer solutions to $x^2 - y^2 = 2024$.

Ans: $(x, y) = (\pm 507, \pm 505), (\pm 255, \pm 251), (\pm 57, \pm 35), (\pm 45, \pm 1)$ — all sign combinations, 16 solutions in total.